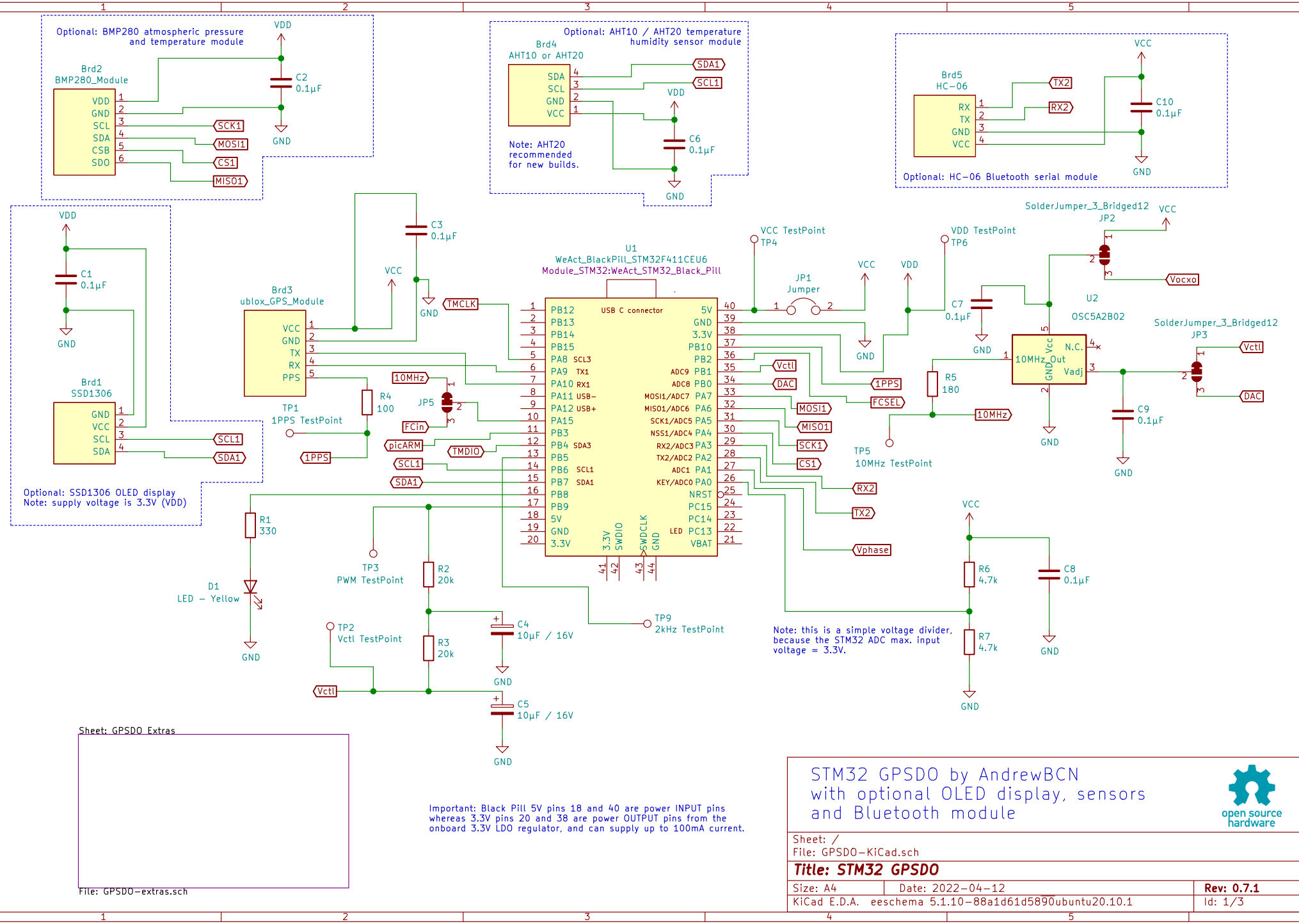
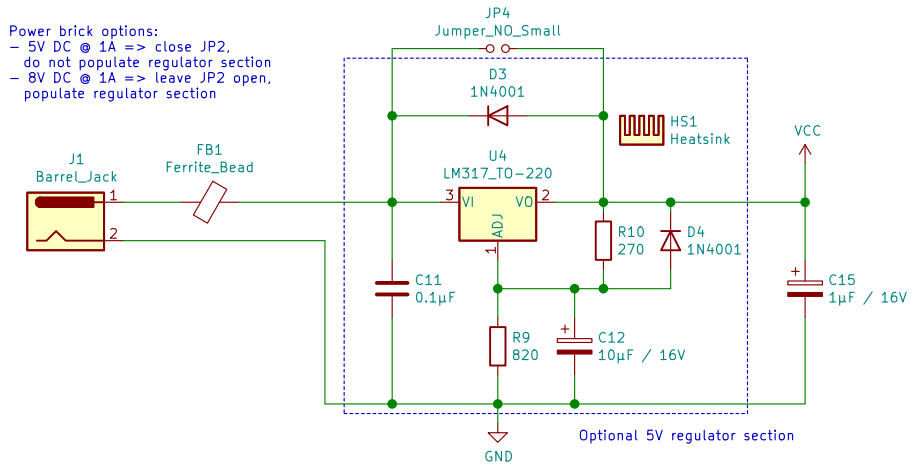
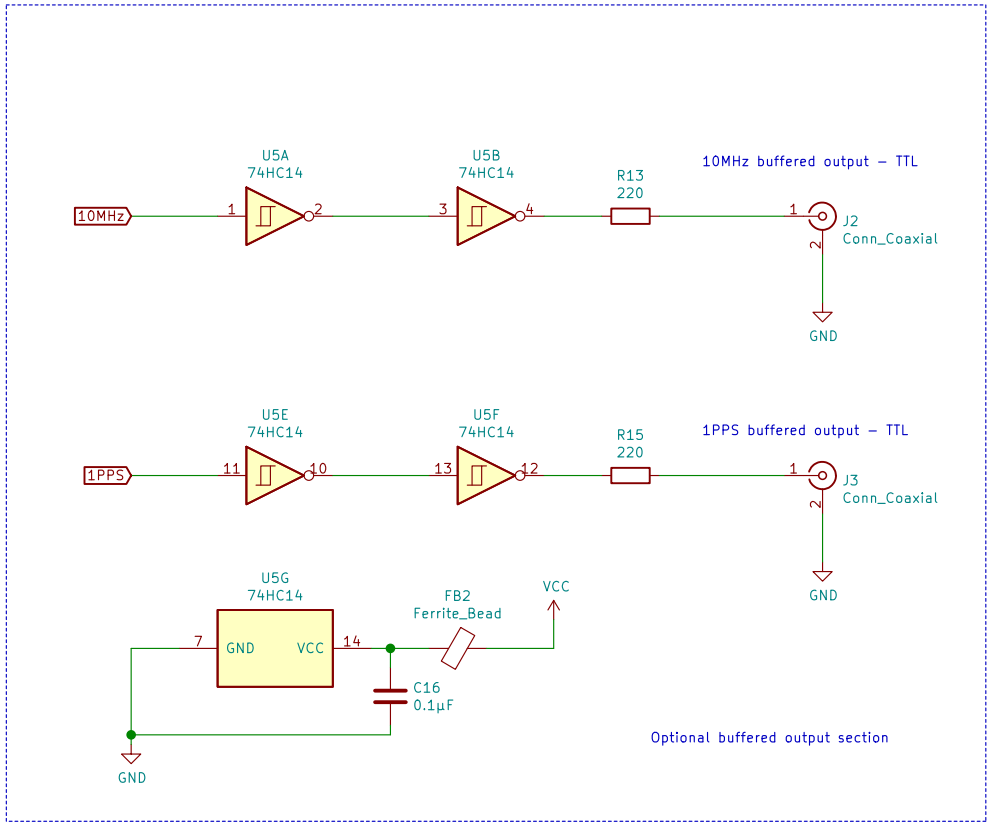
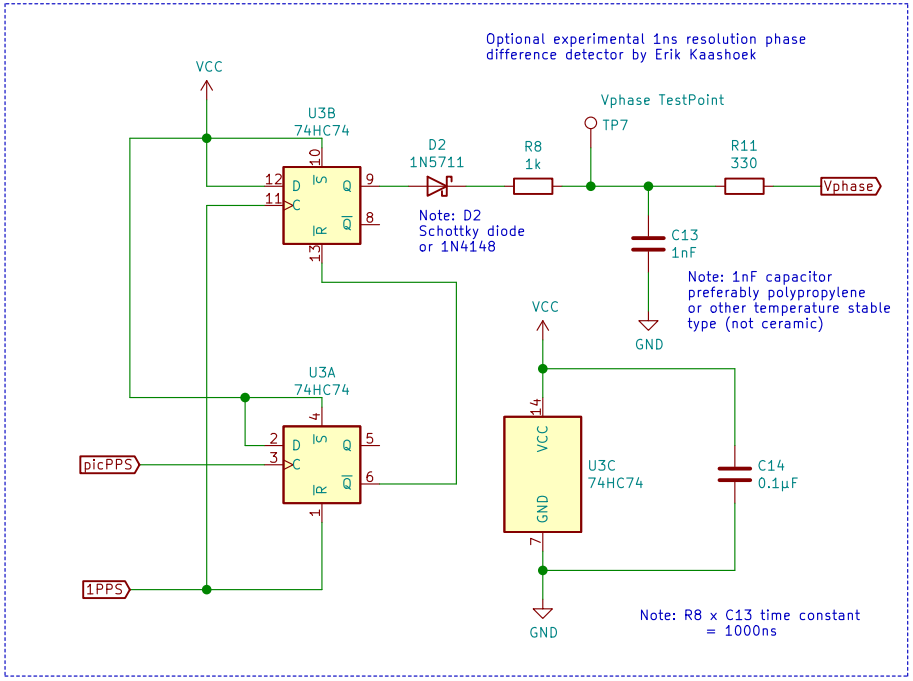

Optional: BMP280 atmospheric pressure and temperature module
Brd2
BMP280_Module
VDD
GND
SCL
SDA
CSB
SDO
SCK1
MOSI1
CS1
MISO1
C2
0.1µF
Optional: AHT10 / AHT20 temperature humidity sensor module
Brd4
AHT10 or AHT20
SDA1
SCL1
C6
0.1µF
Note: AHT20 recommended for new builds.
Optional: HC-06 Bluetooth serial module
Brd5
HC-06
TX
RX
TX2
RX2
C10
0.1µF
Optional: SSD1306 OLED display
Note: supply voltage is 3.3V (VDD)
Brd1
SSD1306
C1
0.1µF
Brd3
ublox_GPS_Module
C3
0.1µF
TMCLK
U1
WeAct_BlackPill_STM32F411CEU6
Module_STM32:WeAct_STM32_Black_Pill
USB C connector
PB12
PB13
PB14
PB15
PA8 SCL3
PA9 TX1
PA10 RX1
PA11 USB-
PA12 USB+
PA15
PB3
PB4 SDA3
PB5
PB6 SCL1
PB7 SDA1
PB8
PB9
5V
GND
3.3V
ADC9 PB1
ADC8 PB0
MOSI1/ADC7 PA7
MISO1/ADC6 PA6
SCK1/ADC5 PA5
NSS1/ADC4 PA4
RX2/ADC3 PA3
TX2/ADC2 PA2
ADC1 PA1
KEY/ADC0 PA0
NRST
PC15
PC14
LED PC13
VBAT
SWDIO
SWDCLK
10MHz
JP5
FCin
picARM
TMDIO
SCL1
SDA1
R4
100
TP1
1PPS TestPoint
1PPS
VCC TestPoint
TP4
JP1
Jumper
VDD TestPoint
TP6
C7
0.1µF
R5
180
TP5
10MHz TestPoint
FCSEL
Vctl
DAC
MOSI1
MISO1
SCK1
CS1
RX2
TX2
Vphase
SolderJumper_3_Bridged12
JP2
Vocxo
U2
OSC5A2B02
10MHz_Out
N.C.
Vadj
C9
0.1µF
SolderJumper_3_Bridged12
JP3
R1
330
D1
LED - Yellow
TP3
PWM TestPoint
R2
20k
R3
20k
C4
10µF / 16V
C5
10µF / 16V
TP2
Vctl TestPoint
TP9
2kHz TestPoint
R6
4.7k
R7
4.7k
C8
0.1µF
Note: this is a simple voltage divider, because the STM32 ADC max. input voltage = 3.3V.
Important: Black Pill 5V pins 18 and 40 are power INPUT pins whereas 3.3V pins 20 and 38 are power OUTPUT pins from the onboard 3.3V LDO regulator, and can supply up to 100mA current.
Sheet: GPSDO Extras
File: GPSDO-extras.sch
STM32 GPSDO by AndrewBCN
with optional OLED display, sensors
and Bluetooth module
open source hardware
Sheet: /
File: GPSDO-KiCad.sch
Title: STM32 GPSDO
Size: A4
Date: 2022-04-12
Rev: 0.7.1
KiCad E.D.A. eeschema 5.1.10-88a1d61d5890ubuntu20.10.1
Id: 1/3







Sheet: GPSDO Extras 2

File: GPSDO-extras2.sch

H1 MountingHole
H2 MountingHole
H3 MountingHole
H4 MountingHole
H5 MountingHole

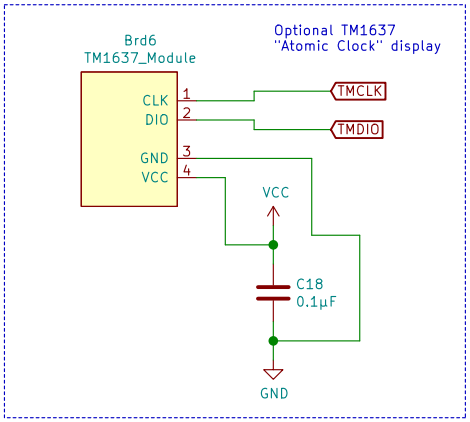
STM32 GPSDO by AndrewBCN
Power supply, output buffers and 1ns resolution phase detector

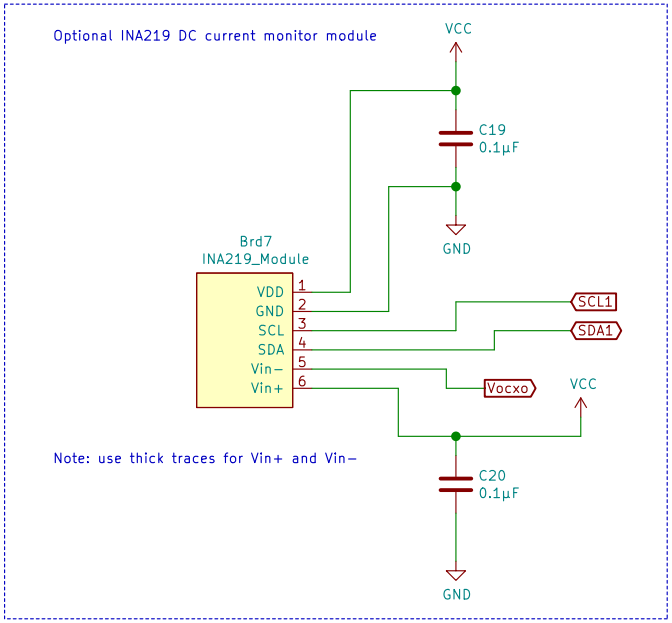
open source hardware

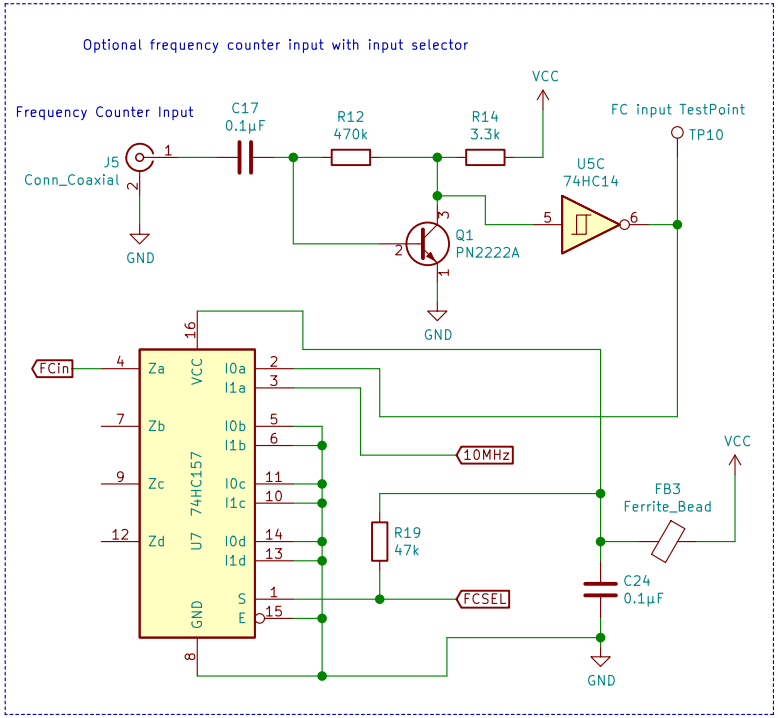
Sheet: /GPSDO Extras/
File: GPSDO-extras.sch

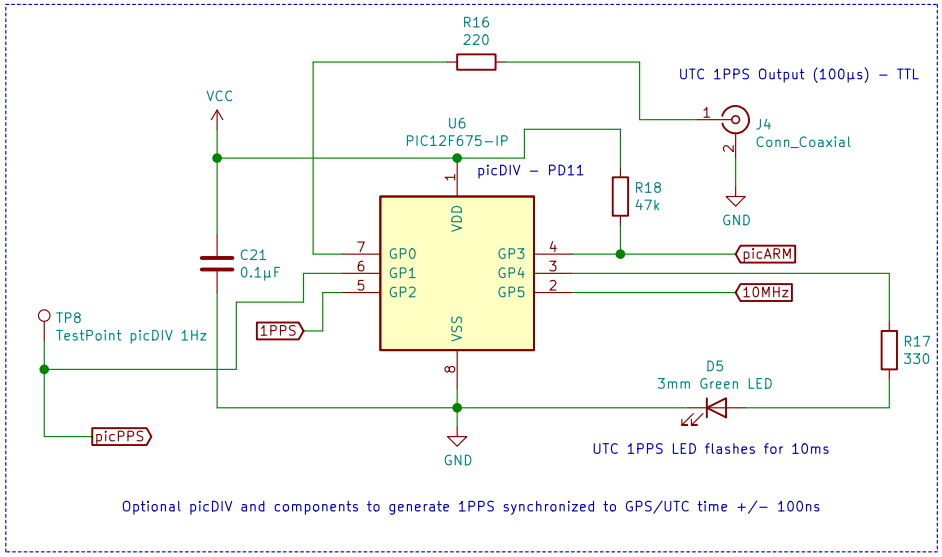
Title: STM32 GPSDO

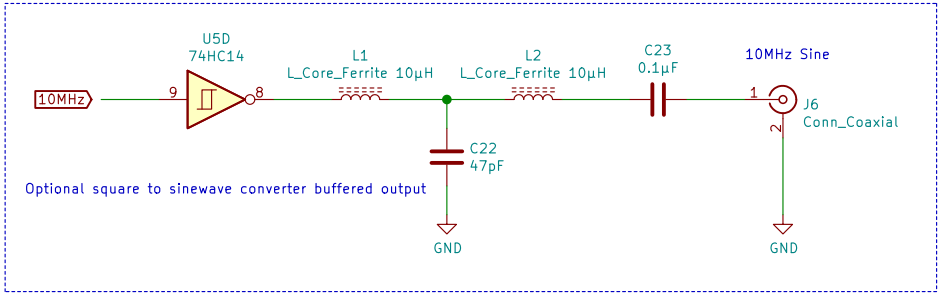
Size: A4	Date: 2022-04-12	**Rev: 0.7.1**
KiCad E.D.A.	eeschema 5.1.10-88a1d61d5890ubuntu20.10.1	Id: 2/3











STM32 GPSDO by AndrewBCN
Optional 12−bit DAC module
Optional OCXO DC current monitor module

open source hardware

Sheet: /GPSDO Extras/GPSDO Extras 2/
File: GPSDO−extras2.sch

Title: STM32 GPSDO

Size: A4	Date: 2022−04−12	**Rev: 0.7.1**
KiCad E.D.A.	eeschema 5.1.10−88a1d61d58~ubuntu20.10.1	Id: 3/3